

Your taste, read from the models

*This is a portrait of your listening drawn **not** by counting plays, but by taking apart the trained “next-track” model and asking what it had to learn about you in order to predict what you play next. Several independent methods are used, and — reassuringly — they point the same way. Raw play statistics appear only near the end, as a cross-check. Exploratory and affectionate, not a verdict.*

A note on the mathematics (kept gentle)

You don’t need machine-learning background for this, only a little linear algebra, probability and the idea of a dynamical system. The objects we use:

- **Tracks are points in space.** The model turns every track into a vector in a learned **192-dimensional “sound space.”** A listening session is then a *path* through that space.
- **The predictor is a recurrence.** Reading a session one track at a time, it keeps a running memory h_t and updates it as

$$h_t = (1 - z_t) \odot n_t + z_t \odot h_{t-1},$$

i.e. at each step it interpolates, coordinate by coordinate ($\odot$ is elementwise), between the old memory h_{t-1} and a fresh candidate n_t , using a learned **keep-fraction** $z_t \in (0, 1)$ (the “update gate”). This is the whole mechanism we later inspect.

- **To read the memory, we factor it.** We fit a dictionary D and sparse codes c so that $h \approx Dc$ with only a handful of nonzero entries (a *sparse autoencoder*). Each dictionary column (“atom”) then behaves like one interpretable concept — almost always a specific artist or genre.
- **Flow is a matrix and a graph.** The co-occurrence (“Markov”) model is just the empirical transition matrix $P(\text{next} \mid \text{current})$; a genre’s **stay-rate** is its diagonal entry. Treating genres as a weighted directed graph, **PageRank** (the stationary distribution of a random walk) finds *hubs* and **betweenness** (fraction of shortest paths through a node) finds *gateways*.
- **Sessions are clustered** with k-means (each session $\rightarrow$ its average vector $\rightarrow$ grouped by nearest centre), and **attractors** are found by *iterating the map* (seed a point, predict the next, jump to the nearest real track, repeat) and watching where the orbit settles.

Everything below is read off these constructions — from inside the model.

1. A map of your listening

Represent each session by its average point in the sound space and cluster them. Six clear **modes** emerge; the shaded “terrain” below is their density — the valleys are where your sessions pile up.

- **Two electronic modes, ~60% of everything.** A club/**UK-garage** mode (Four Tet, Joy Orbison, Aphex Twin, Daphni — 31%) and a **big-beat/trip-hop** mode (The Avalanches, Underworld, Massive Attack, Air — 29%).

**Your taste as a landscape — each dot is one listening session;
valleys (dark) are the modes you settle into**

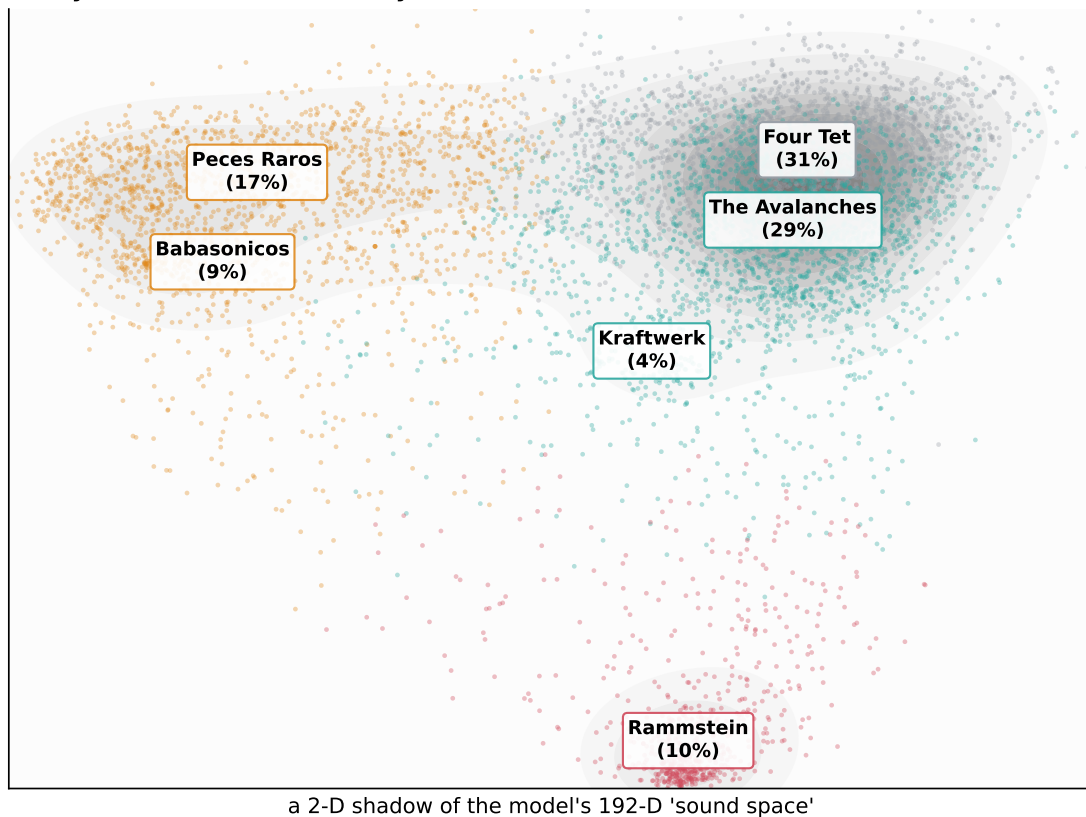


Figure 1: Each dot is one session; darker valleys are the modes you settle into. Argentine music (orange) and the two electronic modes (teal) are large and distinct; the Rammstein/metal basin (red) sits off on its own.

- **Argentine music splits in two:** a contemporary-**indie** mode (Peces Raros, Laika Perra Rusa — 17%) and a **canon** mode (Babasonicos, Gustavo Cerati — 9%).
- **A metal/rock mode (Rammstein, 10%) — and it’s the *short* one:** ~9 tracks per session versus ~14 elsewhere. You binge metal in brief, self-contained bursts.
- **A small kraut/ambient mode (Kraftwerk, Nicolás Jaar — 4%).**

2. What the model considers central

Each concept gets as many dictionary atoms as the model needs to represent it, so **atom-count is the model’s own measure of importance** — independent of how often you played something.

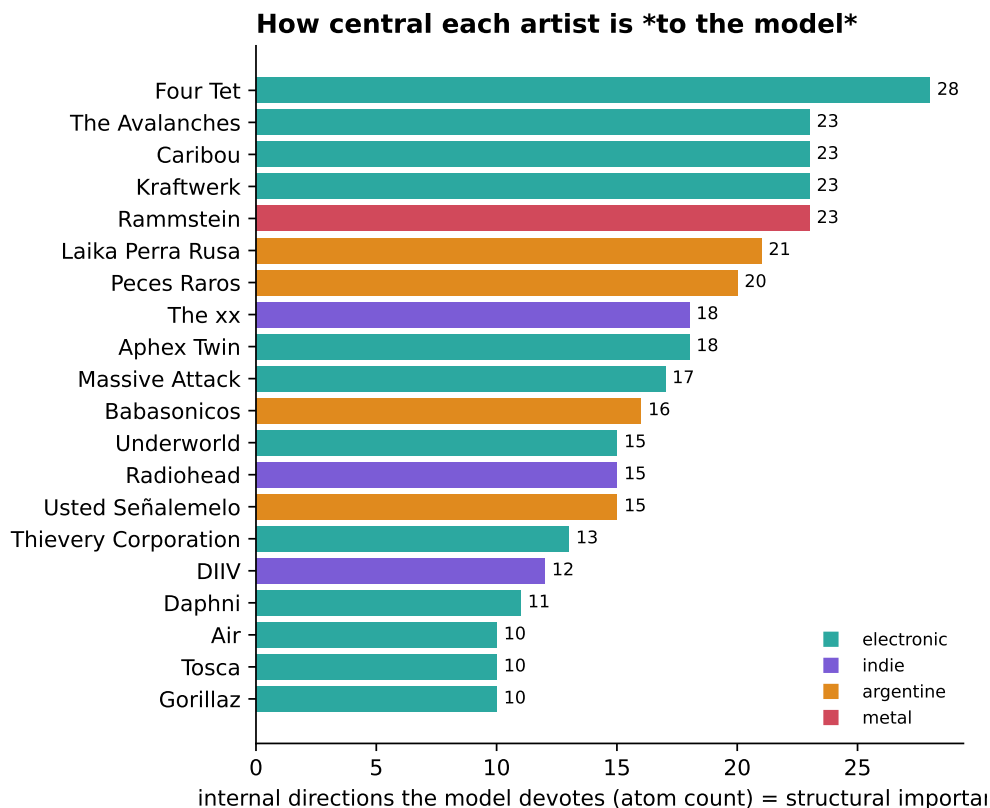


Figure 2: Artists ranked by how many internal directions the model devotes to them.

Four Tet tops the model even though it isn’t your most-played — it takes the most distinct directions to capture. And a result in itself: pushing to **500 atoms** still yields only **~25 artists and ~13 genres**. But that “small cast” must be read carefully — see the next section.

The other 99.6% — your breadth (what the model *can’t* see)

The “small cast” is easy to misread. It is **not** how many artists you listen to; it is how many the model can build *structure* for. By construction the sparse autoencoder only grows an atom for a concept that recurs often enough to help predict the next track — a one-off artist can never earn one. And your listening is mostly one-offs:

You’ve played **5,594 artists**, but the distribution is extreme (Gini **0.88**): **42% you’ve played exactly once, 56% no more than twice** — the median artist just twice. Your top 25 — the

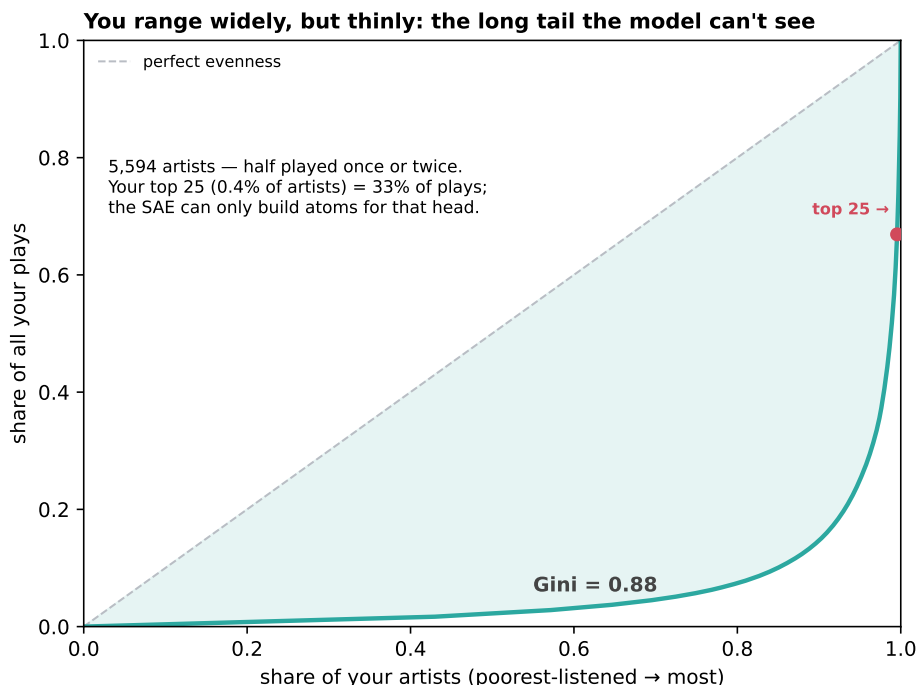


Figure 3: You range very wide but thinly. Cumulative plays vs. cumulative artists (a Lorenz curve): the far the bow bends from the diagonal, the more concentrated. The red dot marks your top 25 — the model’s entire “cast”.

model’s whole cast — are **0.4% of your artists yet a third of all plays**. So the honest picture is the opposite of narrow: you are a **wide-amplitude explorer** with a small, heavy core. Everything else in this document describes that core, because repetition is the only thing the model can reason about; the vast, thin tail of everything-you-tried-once is just as real, and simply **invisible to this lens**.

3. Valleys and passes — “blocks” vs “bridges”

The single sturdiest finding, because **two unrelated methods produce it independently**. For each genre we compute its Markov **stay-rate** (how often it leads to itself) and, separately, the **size of the jump in the neural memory** $\|\Delta h\|$ when a track of that genre arrives. They line up:

- **Blocks** — valleys you stay in: **alternative metal (stay-rate 0.71)**, Kraftwerk’s dusseldorf-electronic (0.55), Argentine rock/indie (0.51).
- **Bridges** — the passes between valleys: melodic techno (0.21), deep house (0.23), dance pop (0.23), art pop. Their *arrival* forces the biggest memory reorganization (the neural state jumps **1.5×** more at a genre switch than within a genre).

The allegory is exact and it’s the same picture as §1: your taste is a **terrain of valleys** (metal, Argentine rock, Kraftwerk) joined by **electronic mountain passes** (house/techno). The electronic core isn’t just music you play — it’s the *transport network* between everything else.

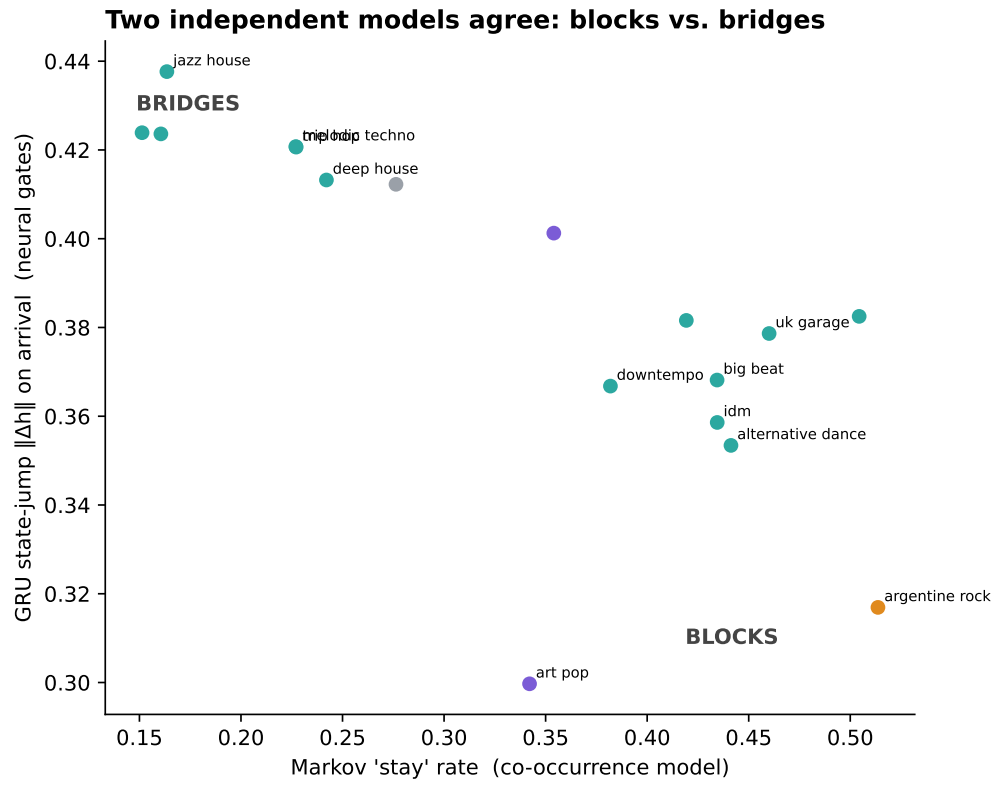


Figure 4: Right/low = you settle in (blocks); left/high = the state reorganizes (bridges). The co-occurrence model and the neural network agree.

4. The flow-grammar

Made explicit as a graph: nodes are genres, arrows are hand-offs, node size is the PageRank “hub” score.

The flow-grammar: how genres hand off (node size = 'hub' score)

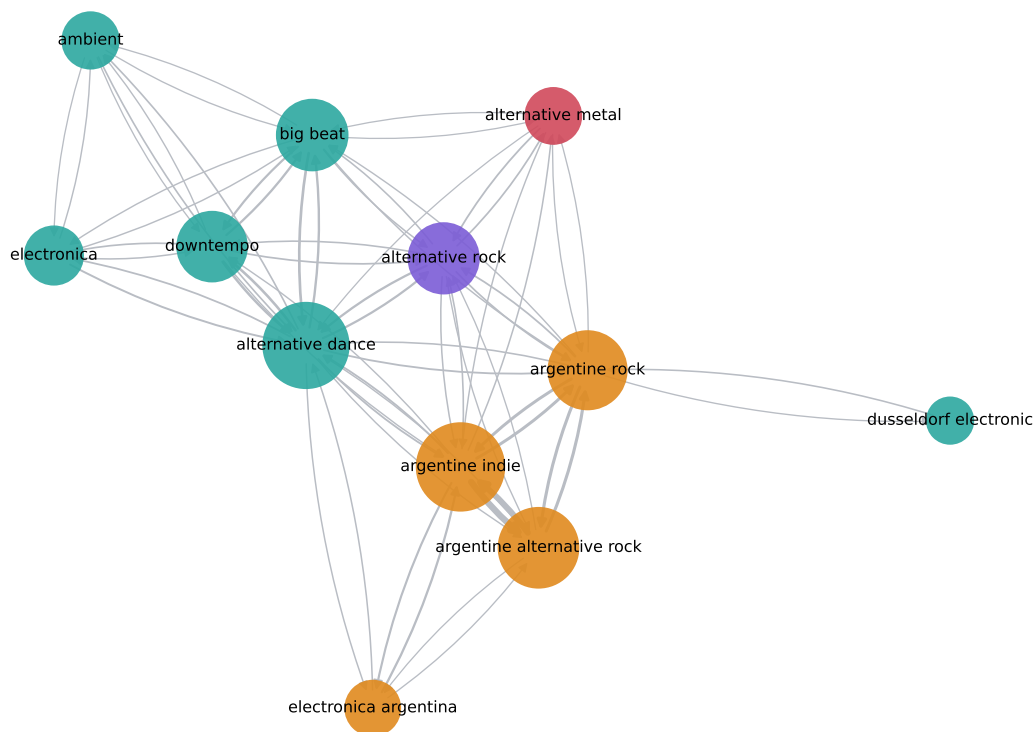


Figure 5: How genres hand off. A tightly interwired Argentine cluster (orange), an interlinked electronic web (teal), and cross-bridges between them.

- **Hubs** (where a wanderer spends time): alternative dance, Argentine indie, downtempo. **Gateways** (bottlenecks between regions, by betweenness): alternative dance and — interestingly — **alternative metal**.
- **Metal book-ends your sessions**: it's your single most common way to both **start** (7.8%) and **end** (7.5%) a session, despite being only 10% of them — you open and close on it, then leave.
- Named cross-bridges: **Kraftwerk** → **Babasonicos** (German electronic into Argentine rock) and **Rammstein** ↔ **Daft Punk** / **Grimes**.

5. Gravity wells — where the model “rolls”

Treat the trained model as a dynamical system: seed one track, let it predict the next, jump to the nearest real track, and repeat 30 times. Where the orbit settles is a **basin of attraction** — literally where a ball would roll on the §1 terrain.

- **Deep wells (inescapable)**: **Kraftwerk** → 100% dusseldorf-electronic, **Aphex Twin** → 100% ambient, **Radiohead** → 100% alternative rock, **Rammstein** → 70% metal — start here and the model stays, mirroring the short, self-contained “block” sessions.

Gravity wells: roll the model forward from one seed → where it settles (green = stays home, red = flows elsewhere)

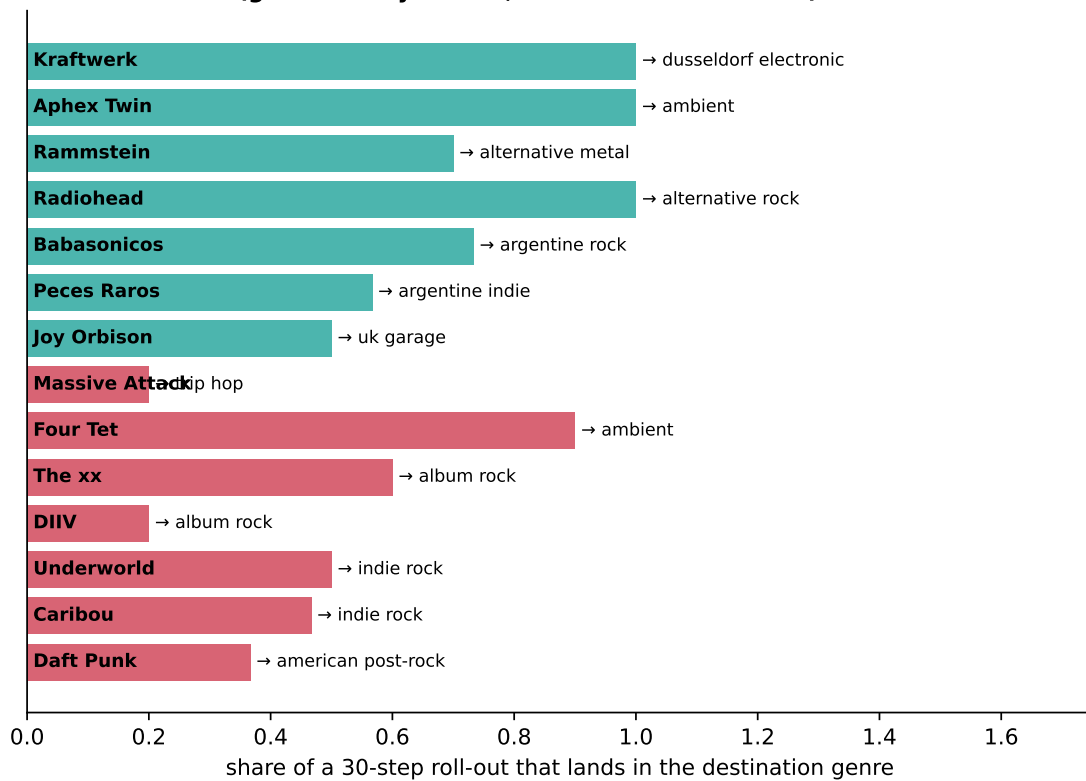


Figure 6: Roll the model forward from one seed. Green = it stays home; red = it flows into another region. Bar length = share of the 30 steps spent in the destination genre.

- **Launch-pads (they flow): Four Tet drains into ambient (90%); The xx, DIIV and Underworld drift toward an album-rock / alternative-rock sink.** So your electronic core is where journeys *begin*; a generic rock/ambient basin is where loosely-guided ones *end up*.

6. Inside the mechanism — how it decides to stay or switch

We can read the update gate z_t exactly (reconstructing the recurrence from the weights reproduces the model to 5×10^{-8}). Two things, one a clean negative:

- **There are no on/off “memory cells.”** The keep-fraction z_t sits at **~0.51 for every one of the 256 coordinates** (range 0.42–0.63) and barely moves — there is no unit that flips *on* for “metal” and *off* for “house.” The memory is **distributed**, not a set of labelled switches.
- **But the state moves 1.5× more at switches** — and that reorganization is what §3’s bridges measure. So the “decide to switch” behaviour is real; it just lives in the whole state’s motion, not in any single interpretable cell.

7. Does it match what you actually do?

As a cross-check against the raw logs (the one place plain counts enter): the model’s highest-salience electronic core is also your **most-played music of the last year**; its “blocks” (Rammstein, Argentine rock) are your biggest **all-time** totals; and the isolated, short Rammstein basin fits your own note that you *have not reached for it in a long time* — the model, learning chronologically, has **demoted the faded phase** even though the lifetime tally still ranks it #1.

In one line

From the inside, your taste is a small, heavy **core** — ~25 artists carrying a third of your plays, floating on a very wide, thin tail of ~5,600 artists you’ve mostly tried once — and that core is arranged as a **terrain**: deep valleys you settle into — Rammstein/industrial metal (short bursts, off on its own), Argentine rock in the Cerati/Babasonicos lineage, Kraftwerk — joined by an **electronic network of passes** (Four Tet, Caribou, Aphex Twin, house/techno) that every method agrees is the true connective structure, over a dream-pop core (DIIV, The xx); and, because the model learns in time, it encodes the **electronic present** you’ve moved into.

Caveats & method: figures are derived from one model on one chronological split; atom-count, stay-rate, $\|\Delta h\|$ and “attractor” are interpretive constructs (a sparse autoencoder over the GRU’s hidden state, the co-occurrence transition matrix, the gate reconstruction, and k-means over session vectors); the neural fingerprint embeds artist names, so concepts skew artist-shaped; only frequent, predictive concepts are represented, not your whole library. Full method: [sae-interpretability-note.pdf](#).